

1. Project overview

The project analyzes eight U.S. equities from January 2015 through December 2023. Monthly adjusted closing prices were converted to returns, and Excel Solver was used to construct a global minimum-variance portfolio, a maximum-Sharpe portfolio, and a sequence of frontier portfolios. The optimized portfolio was benchmarked against SPY and the S&P 500 and then evaluated using market, Fama-French three-factor, and Carhart four-factor regressions.

2. Corrections to the original submission

The original descriptive table displayed standard errors in the row labeled standard deviation. The corrected table reports sample monthly standard deviations. Excel KURT produces excess kurtosis, so normality corresponds to zero. The original market regression reversed the dependent and independent variables; the corrected specification regresses portfolio excess return on S&P 500 excess return. The term CAL is also separated from the Sharpe ratio to avoid interpreting a return coordinate as the slope itself.

3. Descriptive statistics

The corrected statistics show substantial variation in monthly volatility across the selected equities. WBD and MU were among the most volatile series, while JNJ was comparatively low volatility. Jarque-Bera evidence indicates that some individual-stock return distributions depart from normality. These results support the use of covariance-based diversification but also underscore that a normal-return assumption is imperfect.

4. Portfolio optimization

The global minimum-variance portfolio concentrated heavily in JNJ, while the maximum-Sharpe portfolio used positive allocations to CI, JNJ, MU, SHW, and XOM and short positions in Ford, IBM, and WBD. Permitting short sales improved the in-sample frontier but increased implementation complexity. The maximum-Sharpe portfolio had a monthly risk premium of approximately 1.587%, monthly volatility of 6.086%, and monthly Sharpe ratio of 0.261.

5. Benchmark comparison

Annualized performance was approximately 22.4% return and 22.8% volatility for the maximum-Sharpe portfolio, compared with 13.0% return and 13.0% volatility for SPY. The corresponding Sharpe ratios were approximately 0.925 and 0.900. The difference is small relative to the optimized portfolio's higher volatility, concentration, and short exposure, so the project does not establish a compelling economically implementable advantage over the passive benchmark.

6. Factor models

The corrected market regression produced a beta of approximately 0.718 and R-squared of 28.4%. The Fama-French and Carhart models produced R-squared values of 2.1% and 2.6%. The low multifactor explanatory power indicates that the included size, value, and momentum factors captured little of the portfolio's sample variation. Positive sample alpha is interpreted cautiously because the portfolio was optimized and evaluated over the same period.

7. Limitations

The analysis has no out-of-sample validation and does not account for turnover, trading costs, taxes, short-borrow fees, or leverage constraints. Mean-variance optimization is sensitive to estimation error, and the eight-stock universe is narrow. The results are best understood as an educational demonstration of portfolio construction and factor modeling rather than a deployable investment strategy.